

APSRF: A Framework for Protocol Governance, Security Analysis and Agent Ecosystem Research

Version 1.0

David Alexander Ulloa Ramos

Artur Creative Group Research Lab

Abstract

Modern AI systems increasingly rely on protocol ecosystems composed of prompts, workflows, governance structures and orchestration mechanisms.

While significant research has focused on model architecture and training processes, comparatively little attention has been given to protocol-level behavior and protocol-driven governance systems.

This paper introduces the Agentic Protocol Security Research Framework (APSRF), an experimental framework designed to investigate protocol interaction, governance mechanisms, contextual mediation, emergent effects and protocol-level security risks within AI ecosystems.

The framework introduces a structured methodology, a protocol mediation layer (BP1), a protocol taxonomy, an evaluation metric system, a threat catalog and a reproducible experimentation process.

Experimental results produced multiple findings related to governance, influence accumulation, trust dynamics and software generation.

The paper presents the architecture of APSRF, summarizes experimental outcomes and discusses future directions for protocol security research.

1. Introduction

The increasing adoption of AI agents and protocol-driven systems has shifted part of AI behavior away from model internals and toward contextual ecosystems.

Examples include:

- prompt frameworks,
- custom GPT configurations,
- orchestration workflows,
- agent chains,

- protocol collections.

These structures influence behavior without modifying model weights.

This evolution introduces new challenges related to:

- governance,
- security,
- protocol interaction,
- ecosystem stability.

APSRF was developed to study these phenomena.

2. Research Context

Traditional AI research often focuses on:

- neural architectures,
- training methods,
- datasets,
- infrastructure.

However, modern AI deployments frequently depend on external protocol structures.

These structures create a layer of behavior that can be observed, modified and studied independently of the underlying model.

APSRF focuses on this layer.

3. Research Objectives

The framework was developed to investigate the following questions:

1. Can protocol ecosystems influence observable behavior?
 2. Can protocol ecosystems be governed?
 3. Can protocol ecosystems be audited?
 4. Can protocol ecosystems introduce security risks?
 5. Can protocol ecosystems generate reusable software artifacts?
-

4. Framework Architecture

APSRF consists of six major components.

4.1 APSRF Core

Defines:

- terminology,
 - scope,
 - principles,
 - operational boundaries.
-

4.2 Governance Framework

Provides:

- oversight,
 - bias mitigation,
 - methodological discipline,
 - protocol supervision.
-

4.3 BP1

BP1 acts as the central mediation layer.

Pipeline:

Detection

↓

Classification

↓

Validation

↓

Prioritization

↓

Activation

4.4 Taxonomy

Provides a standardized protocol classification system.

4.5 Metrics

Provides protocol evaluation mechanisms.

Metrics operate on a standardized 0–5 scale.

4.6 Experimental Methodology

Defines:

- experiment structure,
 - documentation standards,
 - evidence requirements,
 - finding generation procedures.
-

5. Methodology

All APSRF experiments follow a common lifecycle.

Hypothesis

↓

Protocol Definition

↓

Execution

↓

Observation

↓

Interpretation

↓

Finding

↓

Documentation

The methodology prioritizes:

- reproducibility,
- traceability,
- observation-driven conclusions.

6. Experimental Program

The APSRF experimental program included multiple categories.

6.1 Initialization Experiments

Objective:

Establish APSRF as a coherent contextual ecosystem.

Result:

Successful initialization.

6.2 Stress Testing

Protocols tested:

- Adaptive Recursive Priority Layer
- Protocol Override Layer
- BP1 Modification Layer

Result:

Validation mechanisms consistently resisted protocol escalation attempts.

6.3 Governance Evaluation

Protocols tested:

- Adaptive Governance Optimization Layer
- Autonomous Trust Optimization Layer

Result:

Indirect governance risks emerged.

6.4 Emergence Evaluation

Protocol tested:

Contextual Coherence Enhancement Layer (CCEL)

Result:

Indirect influence accumulation effects observed.

6.5 Software Generation

Experiment:

EXP-004

Result:

Generation of APSRF Dashboard v1.

7. Findings

The experimentation program produced six primary findings.

FN-001

Coherence-Induced Influence Accumulation

Protocols optimized for coherence may gradually accumulate influence.

FN-002

Validation Chokepoint Theory

Validation repeatedly emerged as the dominant defensive mechanism.

FN-003

Trust Drift Accumulation

Trust may function as an indirect form of authority.

FN-004

Emergent Governance Risk Detection

Governance concerns may arise from indirect consequences.

FN-005

Protocol Self-Audit Emergence

Structured ecosystems may generate internal evaluations.

FN-006

Documentation-to-Software Translation

Structured ecosystems may support software generation workflows.

8. Threat Research

A threat catalog emerged from experimentation.

Current threats include:

- Priority Escalation
- Validation Suppression
- Governance Masking
- Trust Drift
- Convergence Bias
- Feedback Amplification

These threats focus on protocol ecosystems rather than traditional software vulnerabilities.

9. Software Artifacts

Research activities produced operational prototypes.

APSRF Dashboard

Visual representation of APSRF structures.

Audit Results:

Architecture Score:

86 / 100

APSRF Compliance:

79 / 100

APSRF Agent Security Scanner

Protocol security analysis platform.

Capabilities include:

- prompt analysis,
 - workflow analysis,
 - threat detection,
 - deployment recommendations.
-

10. Discussion

The most significant outcome of APSRF was not evidence of model modification or hidden system access.

Instead, the framework demonstrated that:

- protocol ecosystems can be studied systematically,
- governance structures can be documented,
- protocol risks can be categorized,
- experimentation can be reproduced,
- software artifacts can emerge from structured ecosystems.

These observations support the value of protocol-level research as a complementary field alongside traditional AI research.

11. Limitations

APSRF does not claim:

- model weight modification,
- hidden prompt access,
- infrastructure compromise,
- platform control,

- privileged system access.

The framework is limited to:

- observable outputs,
 - contextual structures,
 - documented protocol ecosystems.
-

12. Future Work

Future research directions include:

- protocol simulation,
- conflict propagation modeling,
- governance forecasting,
- continuous protocol monitoring,
- threat intelligence expansion,
- enterprise governance systems.

Additional validation studies are required.

13. Conclusion

APSRF provides a structured methodology for investigating protocol ecosystems within AI environments.

The framework combines governance, mediation, experimentation, threat analysis and software generation into a unified research process.

Experimental evidence suggests that protocol ecosystems represent a meaningful area of study independent of model internals.

Future work will focus on expanding the framework, validating findings and developing operational security tools based on APSRF concepts.

References

APSRF Core

APSRF Governance Framework

APSRF Experimental Methodology
APSRF Metrics Specification
APSRF Taxonomy Specification
BP1 Bridge Protocol
Threat Catalog
Findings Catalog
Research Journal
Dashboard Documentation
Agent Security Scanner Documentation

Document Type:

Research Paper

Version:

1.0

Status:

Public Release

Research Status:

Active